Name: Kashinath Sangram Hale

PRN no.: 202501040055

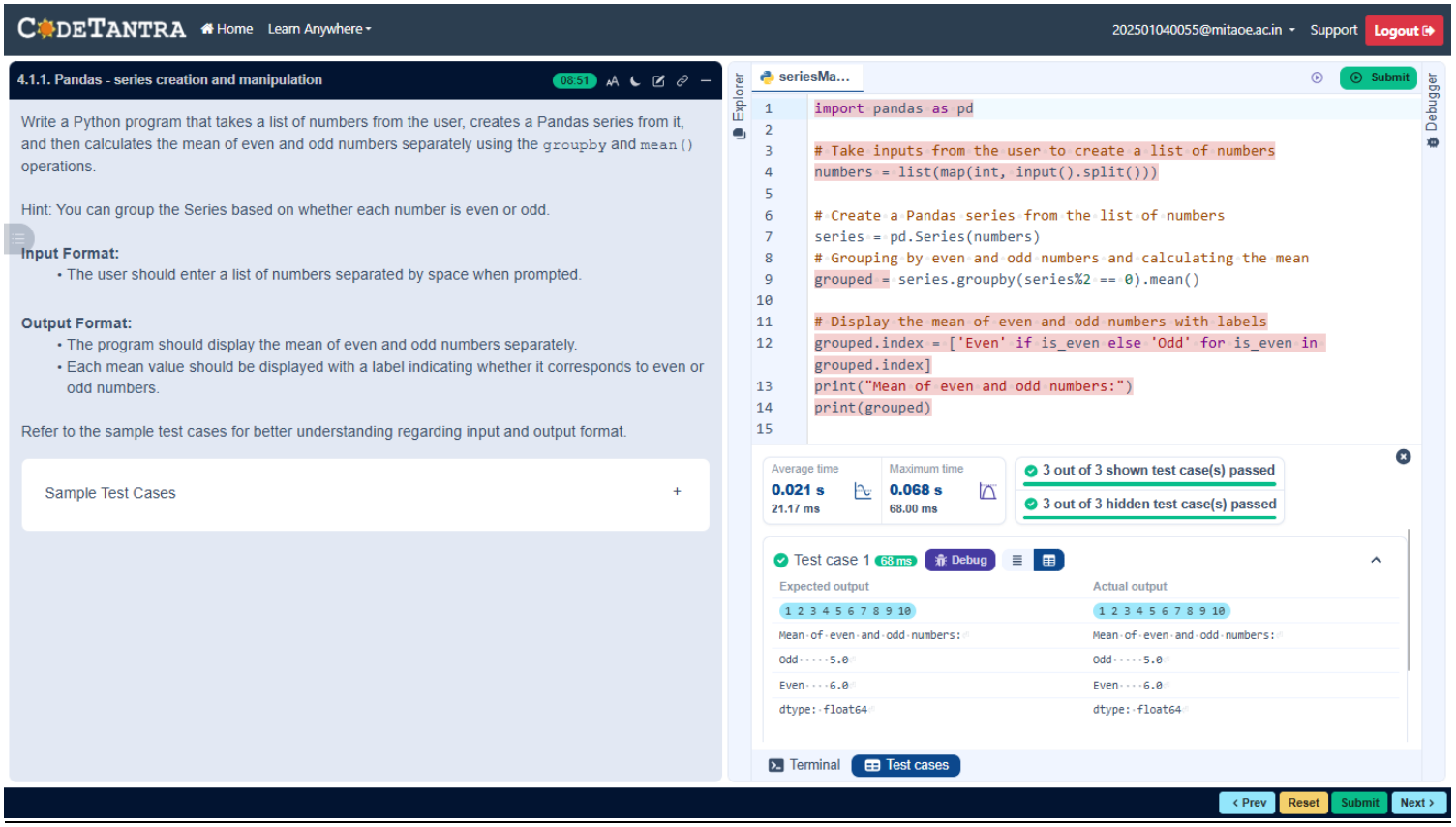
Division: CS-1 Batch:

C13 Course: EDS

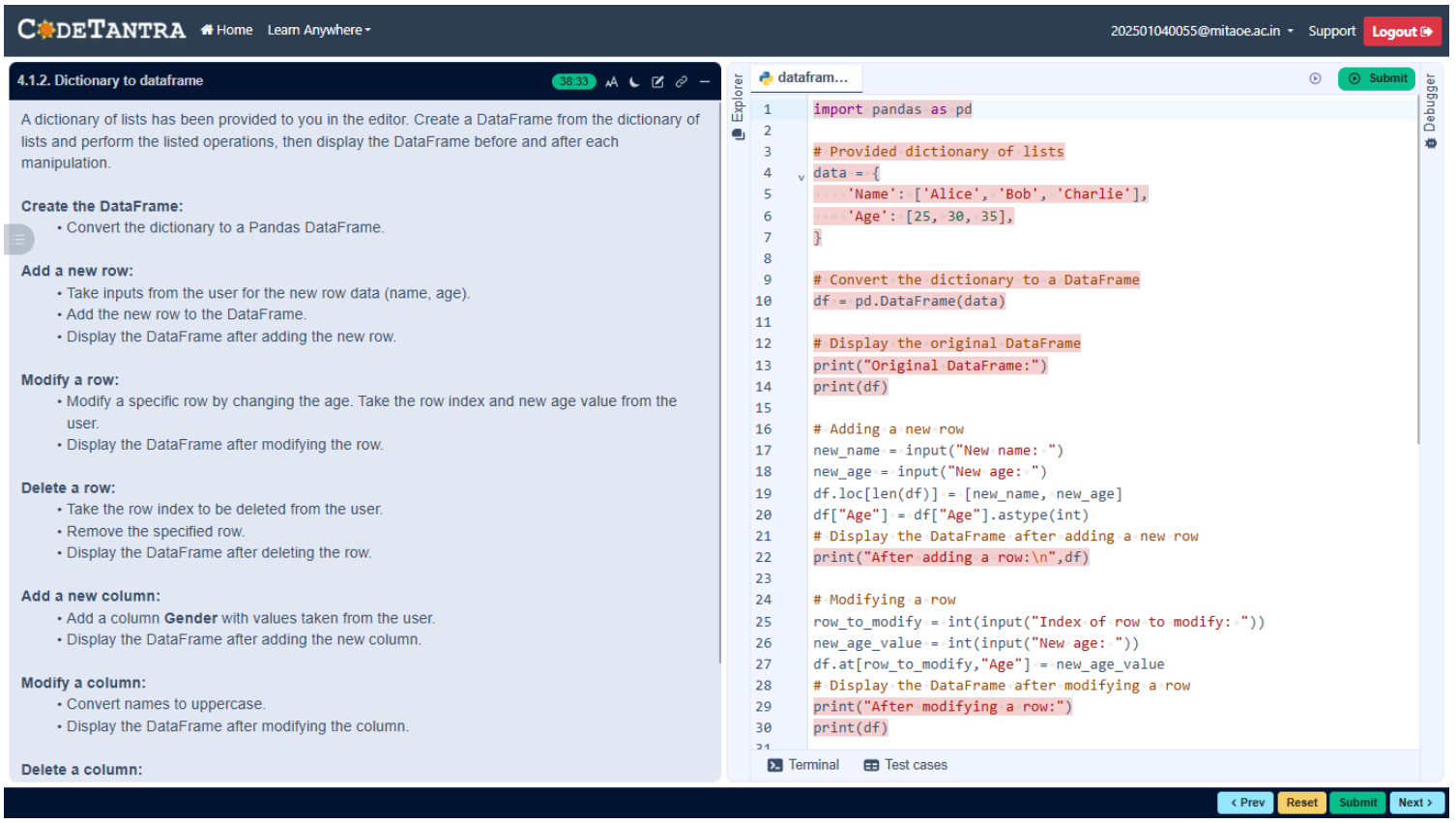
Essential of Data Science Laboratory

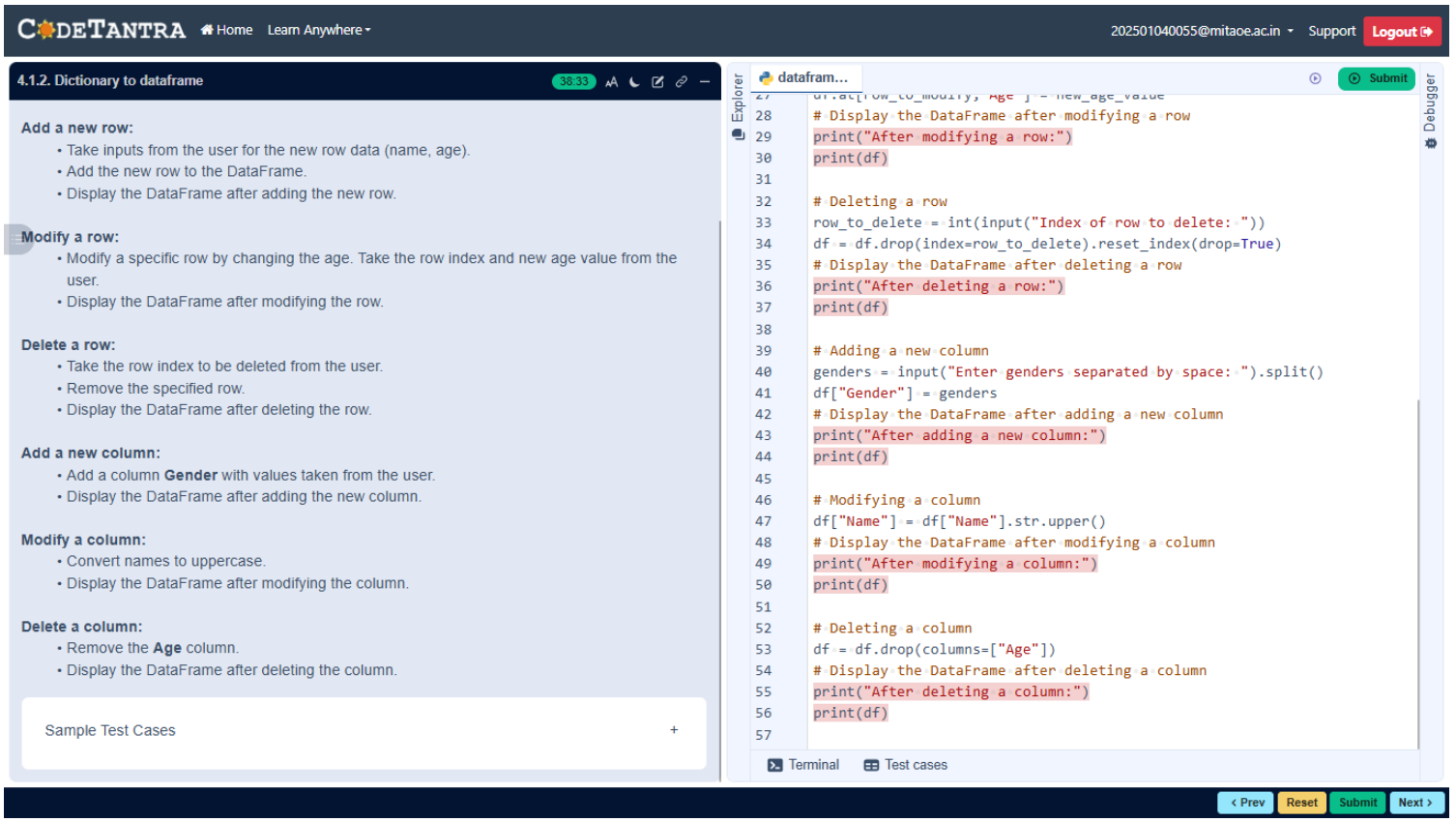
Practical 4:

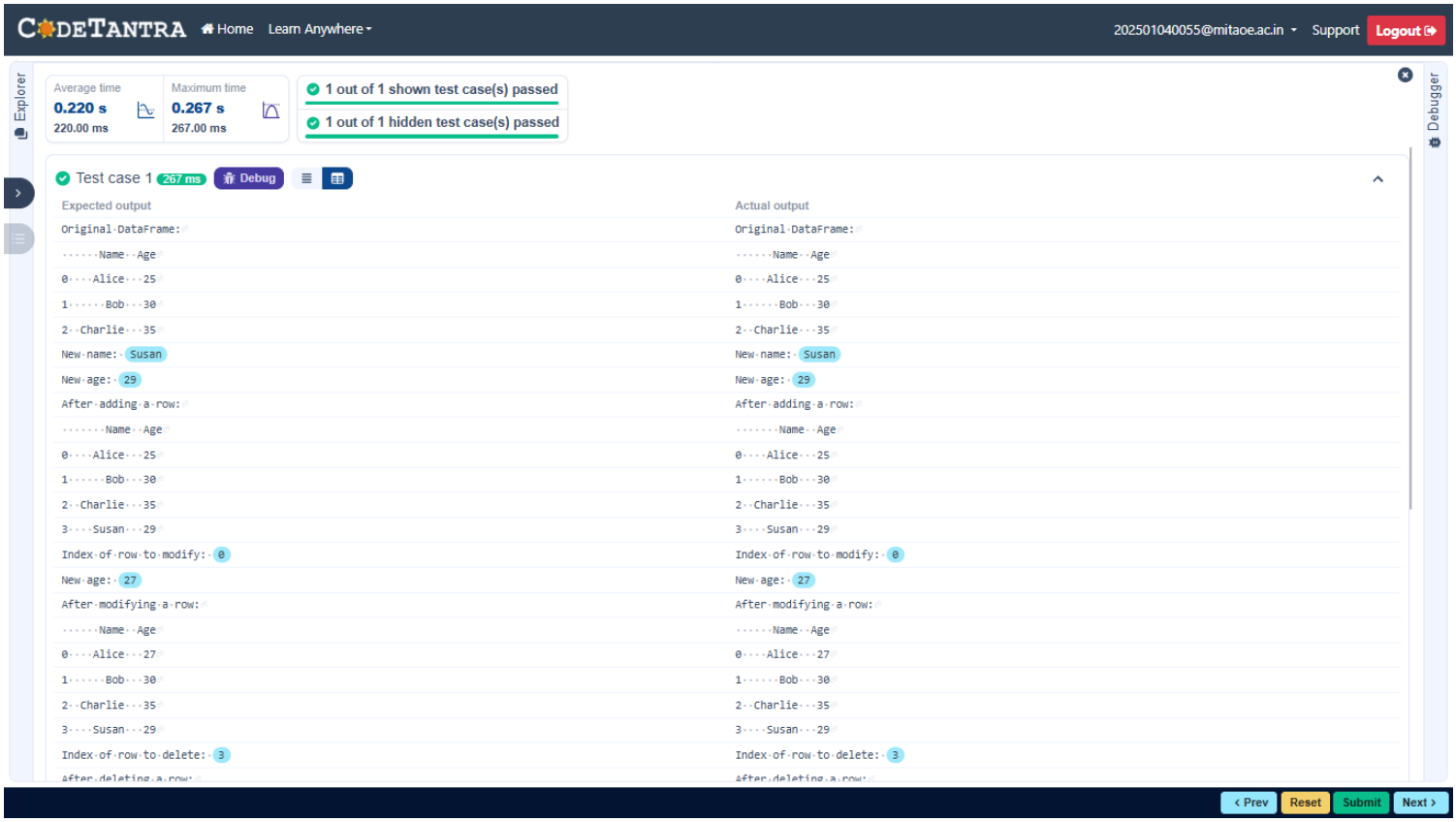
4.1.1 Pandas - Series Creation and Manipulation:



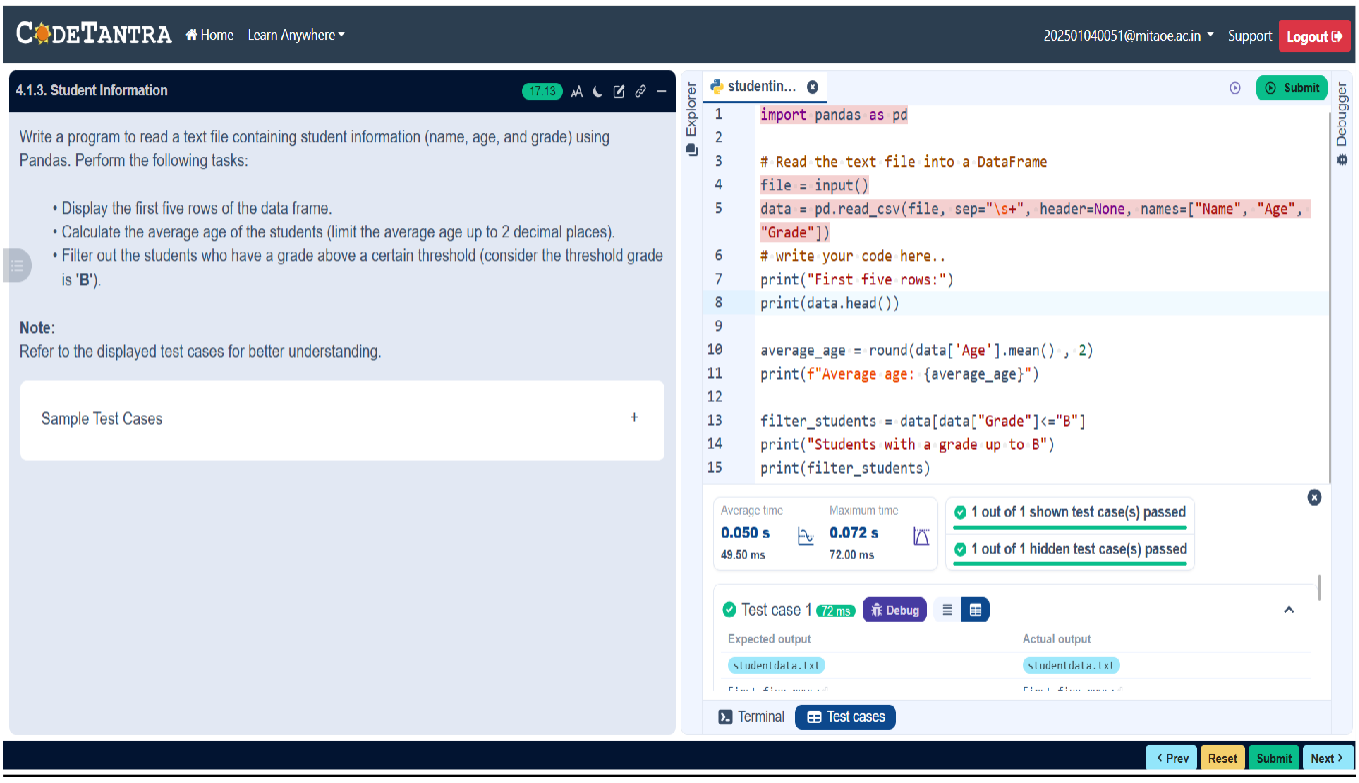
4.1.2 Dictionary to dataframe:



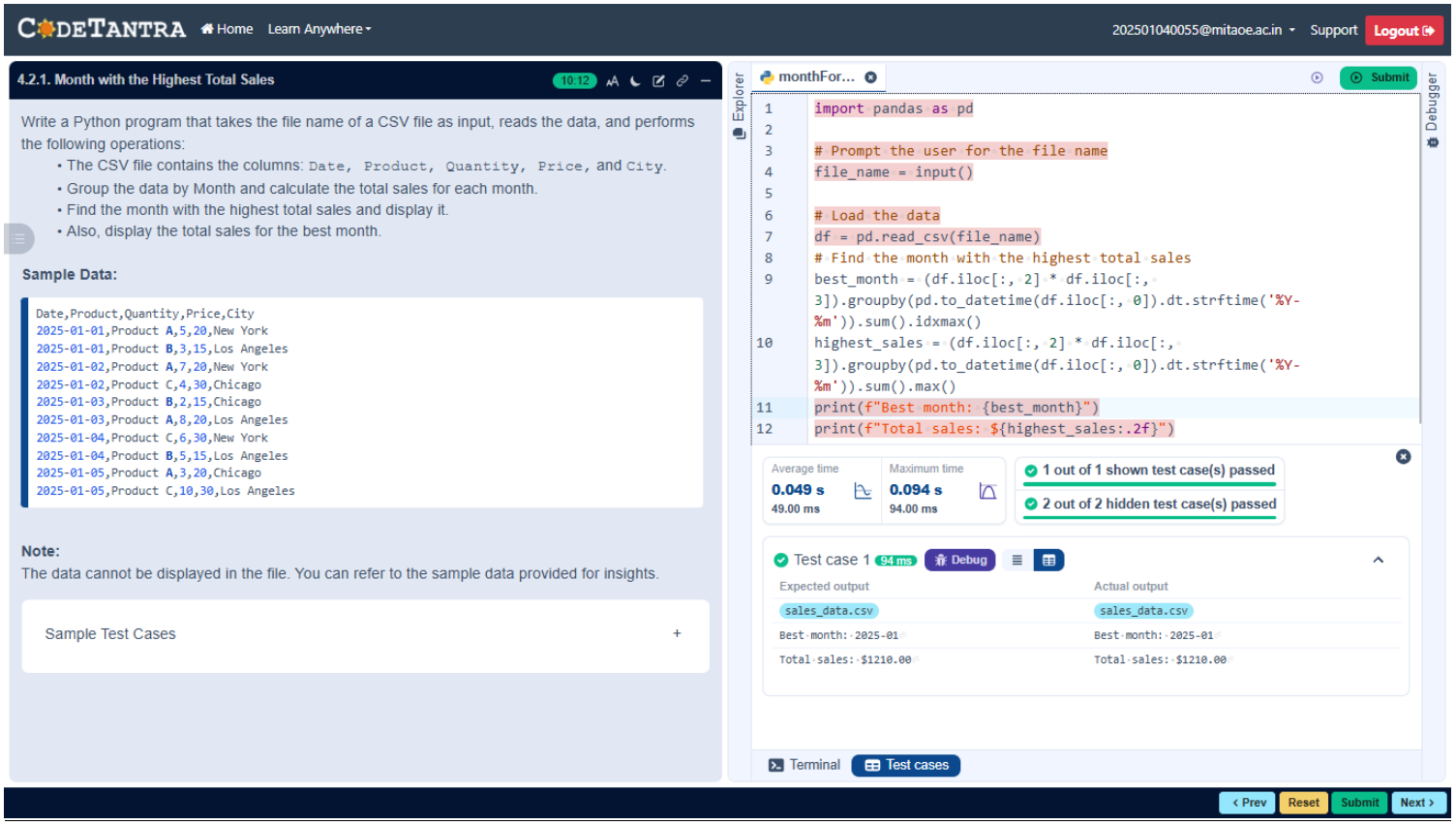




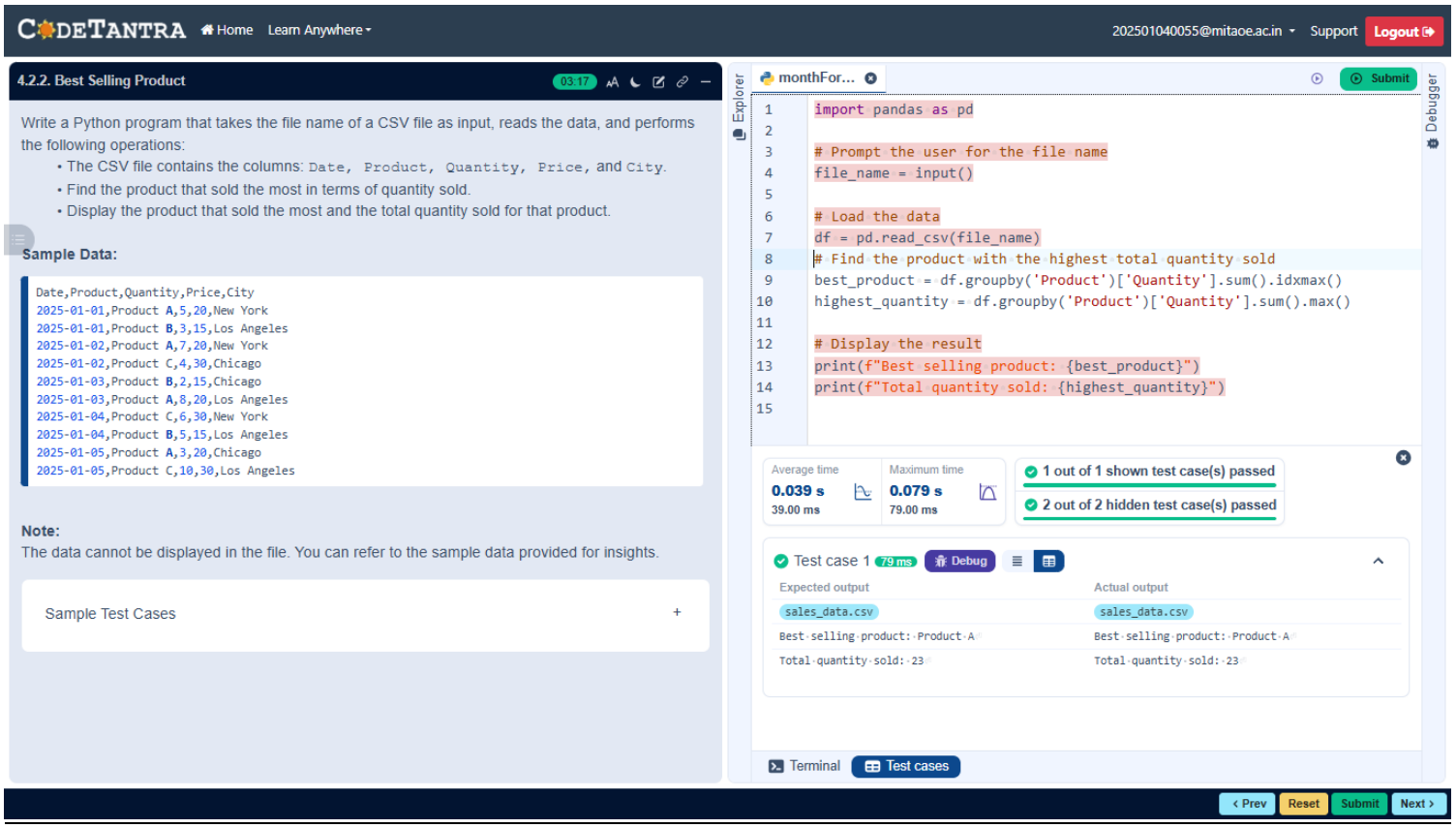
4.1.3 Student Information:



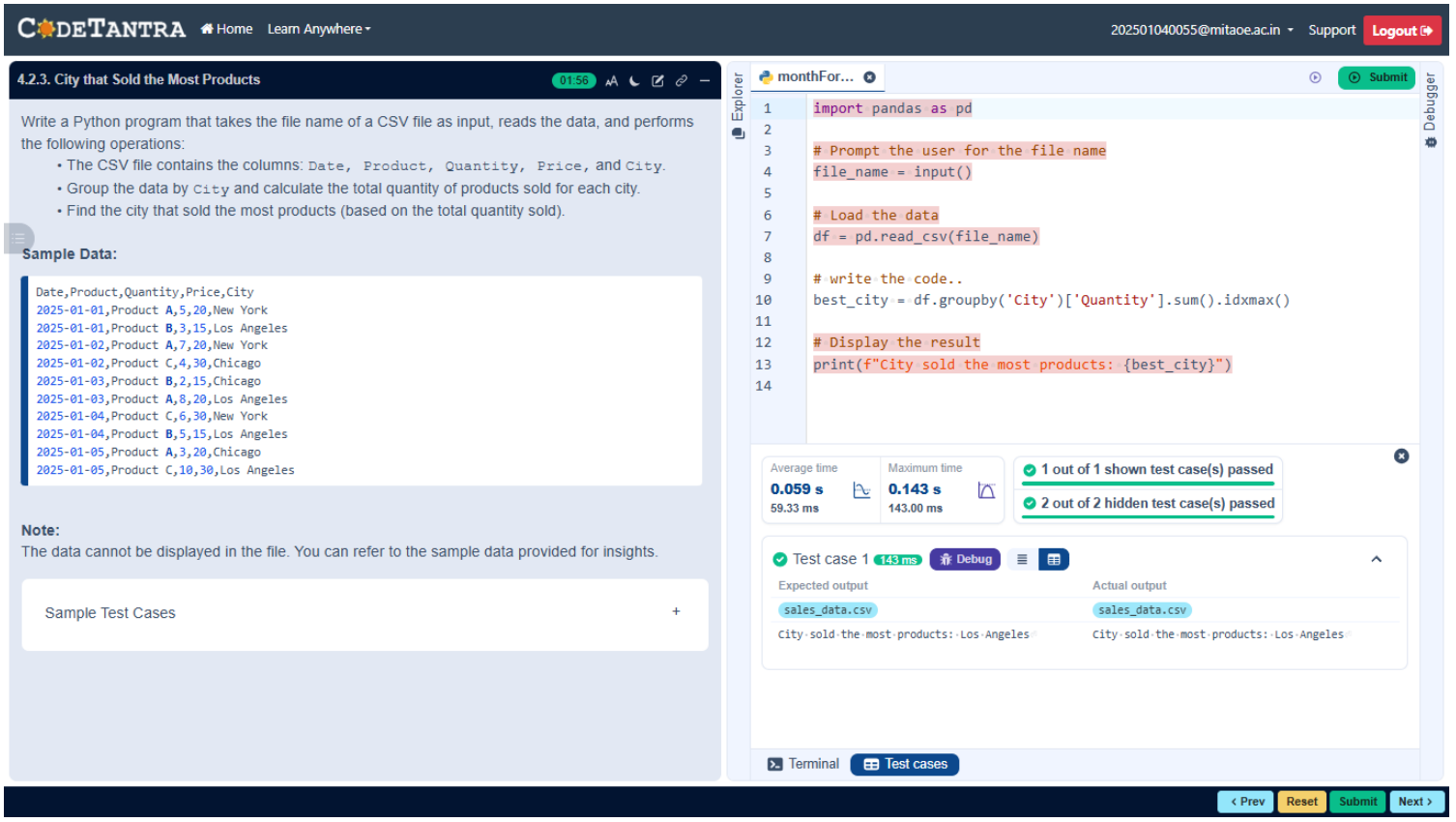
4.2.1 Month with Highest total Sales:



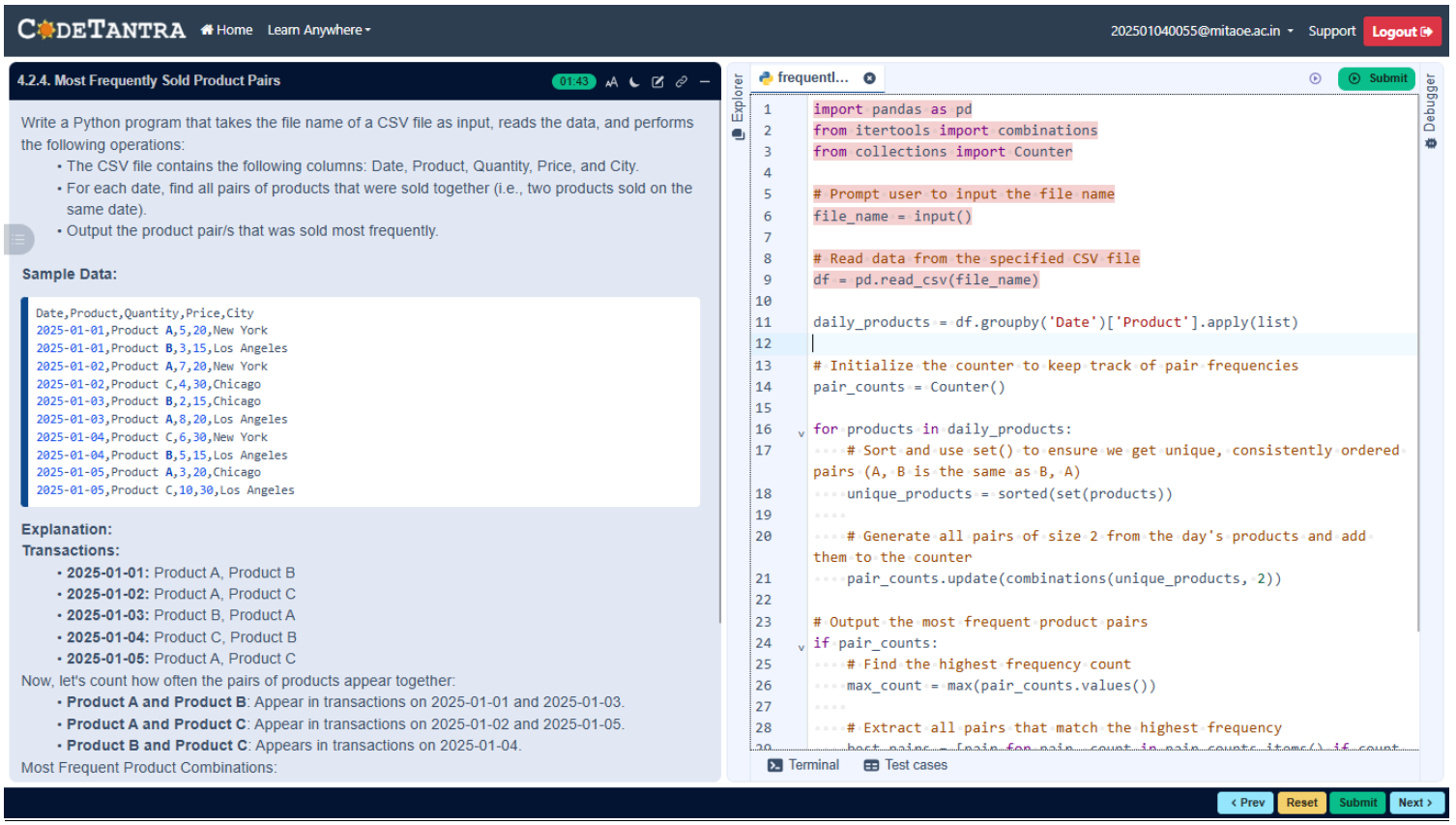
4.2.2 Best Selling Product:

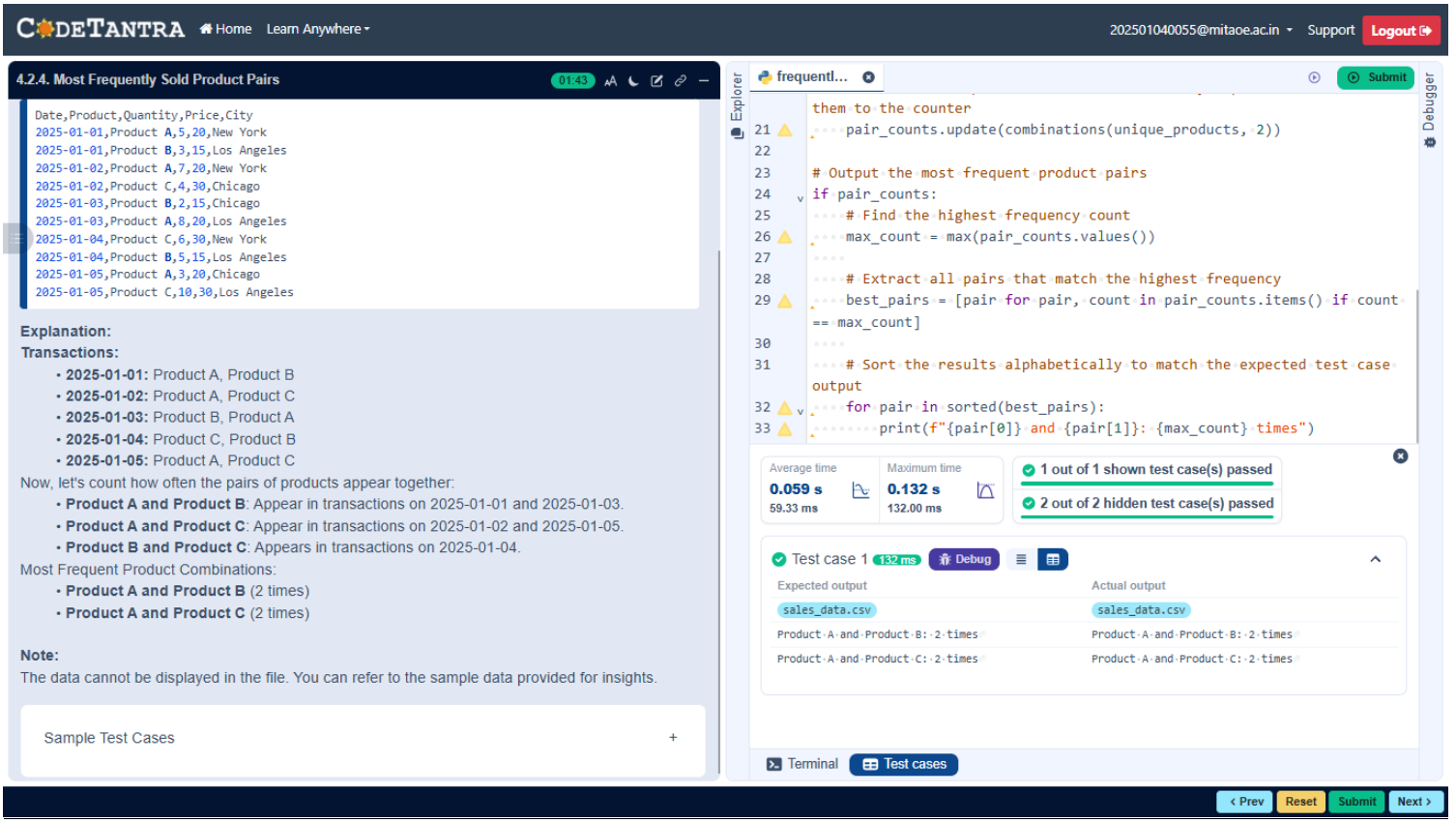


4.2.3 City that Sold Most Products:

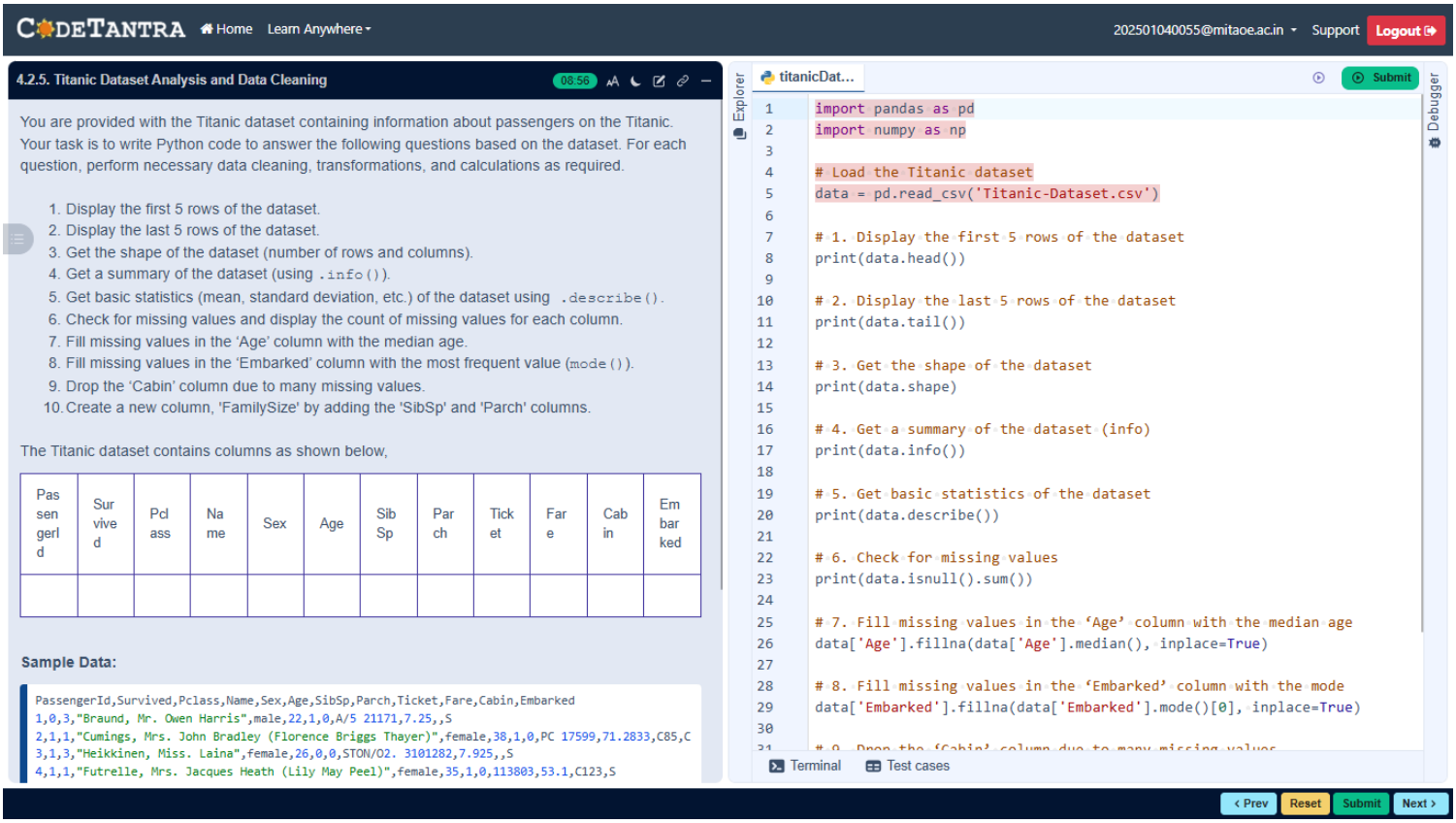


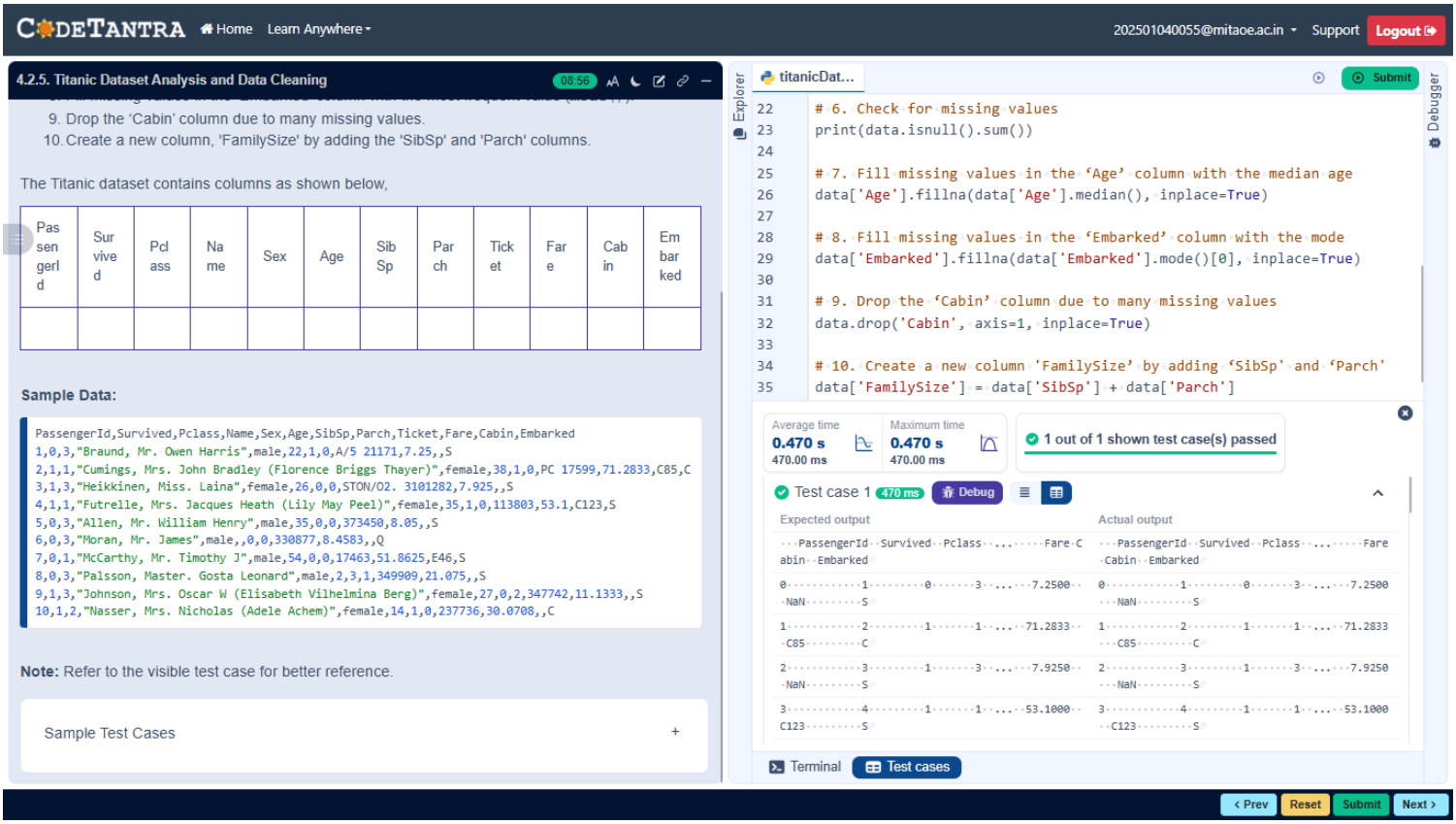
4.2.4 Most Frequently Sold Product Pairs:





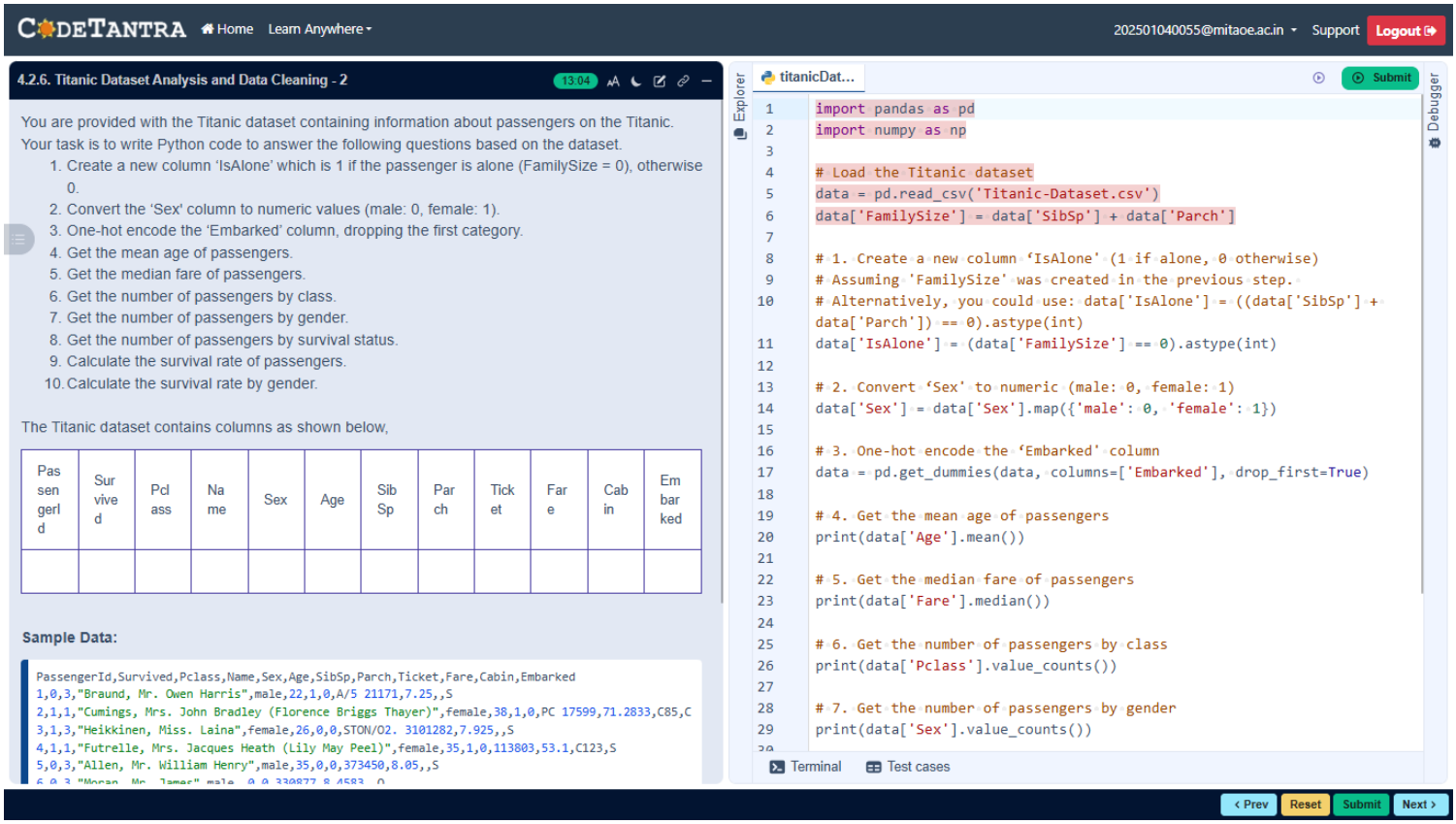
4.2.5 Titanic Dataset Analysis:





4.2.6 Titanic Dataset Analysis:

5



4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

13:04

9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

Passengerld	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases +

titanicDat... Submit

```
27
28  # 7. Get the number of passengers by gender
29  print(data['Sex'].value_counts())
30
31  # 8. Get the number of passengers by survival status
32  print(data['Survived'].value_counts())
33
34  # 9. Calculate the survival rate
35  print(data['Survived'].mean())
36
37  # 10. Calculate the survival rate by gender
38  print(data.groupby('Sex')['Survived'].mean())
```

Average time 0.151 s 151.00 ms | Maximum time 0.151 s 151.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 151 ms Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3 491	3 491
1 216	1 216
2 184	2 184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0 577	0 577
1 314	1 314
Name: Sex, dtype: int64	Name: Sex, dtype: int64

Terminal Test cases

‹ Prev Reset Submit Next ›

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

01:49

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Passengerld	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
```

titanicDat... Submit

```
1   import pandas as pd
2   import numpy as np
3
4   # Load the Titanic dataset
5   data = pd.read_csv('Titanic-Dataset.csv')
6   data['FamilySize'] = data['SibSp'] + data['Parch']
7   data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8   data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10  # 1. Calculate the survival rate by class
11  print(data.groupby('Pclass')['Survived'].mean())
12
13  # 2. Calculate the survival rate by embarked location
14  # Note: pd.get_dummies with drop_first=True usually creates
    'Embarked_Q' and 'Embarked_S'
15  print(data.groupby('Embarked_S')['Survived'].mean())
16
17  # 3. Calculate the survival rate by family size
18  print(data.groupby('FamilySize')['Survived'].mean())
19
20  # 4. Calculate the survival rate by being alone
21  print(data.groupby('IsAlone')['Survived'].mean())
22
23  # 5. Get the average fare by class
24  print(data.groupby('Pclass')['Fare'].mean())
25
26  # 6. Get the average age by class
27  print(data.groupby('Pclass')['Age'].mean())
28
29  # 7. Get the average age by survival status
```

Terminal Test cases

‹ Prev Reset Submit Next ›

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

01:49

9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Passengerld	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases +

titanicDat... Submit

Explorer Debugger

```
print(data.groupby('Pclass')['Age'].mean())

# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())

# 8. Get the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())

# 9. Get the number of survivors by class
print(data[data['Survived'] == 1]['Pclass'].value_counts())

# 10. Get the number of non-survivors by class
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Average time 0.172 s 172.00 ms | Maximum time 0.172 s 172.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 172 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 0.506073	0 0.506073
1 0.336957	1 0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64

Terminal Test cases

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

10:24

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

Passengerld	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
```

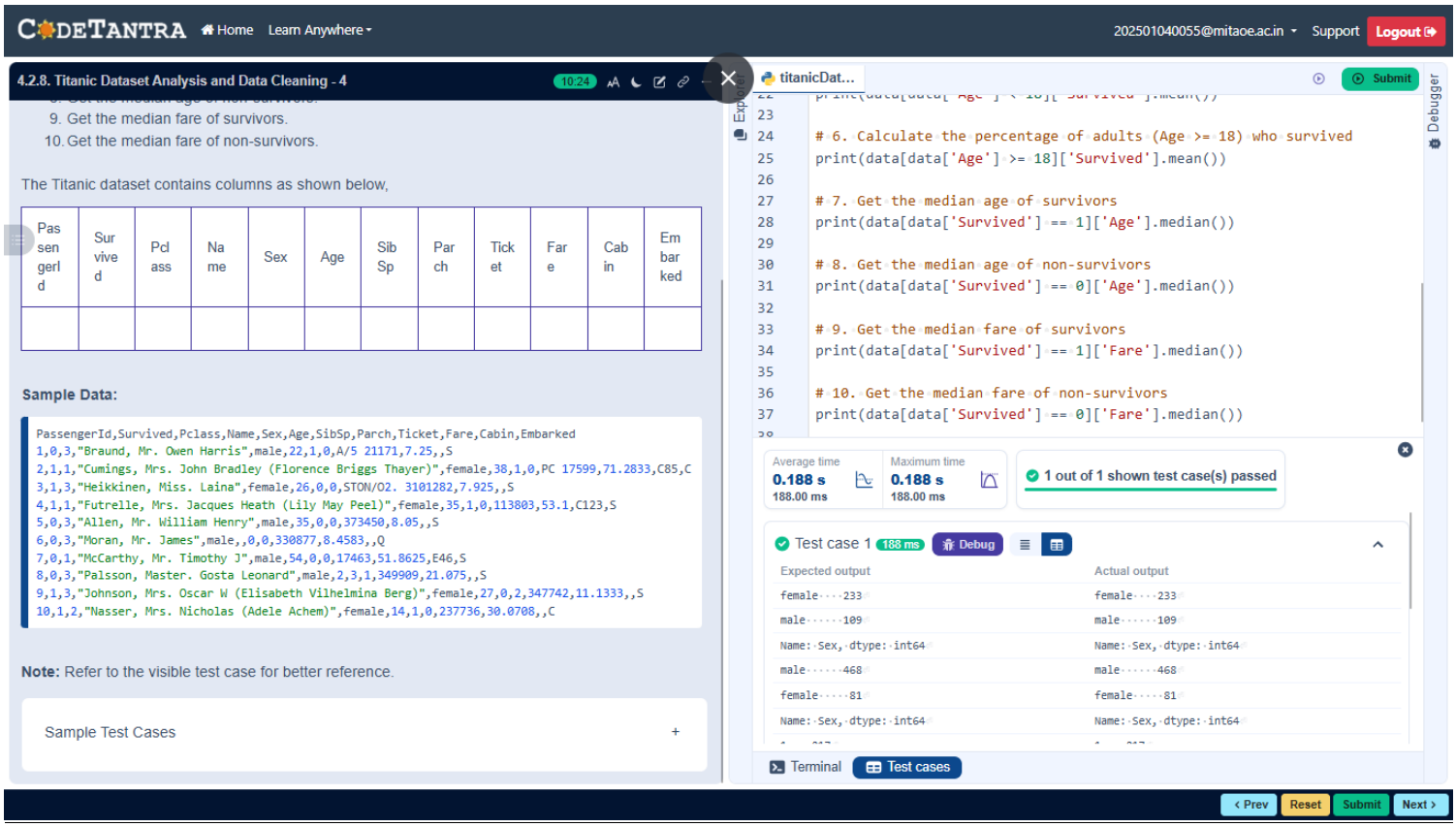
titanicDat... Submit

Explorer Debugger

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# 1. Get the number of survivors by gender
print(data[data['Survived'] == 1]['Sex'].value_counts())

# 2. Get the number of non-survivors by gender
print(data[data['Survived'] == 0]['Sex'].value_counts())

# 3. Get the number of survivors by embarked location
print(data[data['Survived'] == 1]['Embarked_S'].value_counts())

# 4. Get the number of non-survivors by embarked location
print(data[data['Survived'] == 0]['Embarked_S'].value_counts())

# 5. Calculate the percentage of children (Age < 18) who survived
print(data[data['Age'] < 18]['Survived'].mean())

# 6. Calculate the percentage of adults (Age >= 18) who survived
print(data[data['Age'] >= 18]['Survived'].mean())

# 7. Get the median age of survivors
print(data[data['Survived'] == 1]['Age'].median())

# 8. Get the median age of non-survivors
print(data[data['Survived'] == 0]['Age'].median())
```

Terminal Test cases

‹ Prev Reset Submit Next ›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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4
10:24
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.
The Titanic dataset contains columns as shown below,
PassengerId
Survived
Pclass
Name
Sex
Age
SibSp
Parch
Ticket
Fare
Cabin
Embarked
Sample Data:
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
Note: Refer to the visible test case for better reference.
Sample Test Cases
titanicDat...
Submit
Explorer
Debugger
# 6. Calculate the percentage of adults (Age >= 18) who survived
print(data[data['Age'] >= 18]['Survived'].mean())
# 7. Get the median age of survivors
print(data[data['Survived'] == 1]['Age'].median())
# 8. Get the median age of non-survivors
print(data[data['Survived'] == 0]['Age'].median())
# 9. Get the median fare of survivors
print(data[data['Survived'] == 1]['Fare'].median())
# 10. Get the median fare of non-survivors
print(data[data['Survived'] == 0]['Fare'].median())
Average time
0.188 s
188.00 ms
Maximum time
0.188 s
188.00 ms
1 out of 1 shown test case(s) passed
Test case 1
188 ms
Debug
Expected output
Actual output
female 233
male 109
Name: Sex, dtype: int64
male 468
female 81
Name: Sex, dtype: int64
Terminal
Test cases
‹ Prev
Reset
Submit
Next ›